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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CBCS) Degree Examinations
 Summer Semester July 2024

22EVT122 – MODELING AND ANALYSIS OF ELECTRICAL MACHINES

Duration: 3 Hours

Max. Marks:100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **one full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Given $v_a = V_m \sin \omega t$, $v_b = V_m \sin(\omega t + 120)$ and $v_c = V_m \sin(\omega t - 120)$ find V_o , V_d and V_q using power-invariant Park transformation matrix by taking $\theta = \omega t$.	10	L*3	1	3,4
	b) Describe Kron's primitive machine. Derive the voltage and torque equations for this machine.	10	L4	2	3,4
2.	a) Derive the state-space model for shunt- and series-DC motors.	10	L4	2	3,4,5
	b) Elaborate why time-variation of induced emf has the same shape as space-variation of radial component of air-gap flux density.	6	L2	1	3
	c) State the limitations of Kron's primitive machine.	4	L2	2	3
3.	a) Obtain the state-space model of a separately-excited DC motor for a small perturbation.	10	L3	2	3,4,5
	b) A three-phase star-connected balanced voltage source is supplying a three-phase symmetrical load. The load on each phase comprises of an identical series combination of a resistor and a capacitor. Obtain the state equations in $0dq$ frame of reference using power-invariant Park transformation.	10	L3	1	3,4
Unit – II					
4.	a) Determine the voltages induced in d and q axes of rotor and stator of a three-phase induction motor in arbitrary reference frame. Obtain the same in rotor reference frame.	10	L4	3	3,4,5
	b) Derive the torque equation of a three-phase induction motor in stationary reference frame in terms of stator- and rotor-currents. Express in per unit.	10	L4	3	3,4,5
5.	a) The no-load and locked-rotor test results for a three-phase, star-connected, 60-Hz, 2000-hp induction motor with a stator phase resistance of 0.02 ohm are as follows: No-load test: 2300 V input line voltage, 26.55 A line current and 11.617 kW input three-phase power. Locked rotor test: 462.68 V input line voltage, 407.75 A line current and 319.22 kW input three-phase power. Determine the machine equivalent circuit parameters. Also find out self-inductances of stator and rotor.	10	L3	3	3,4
	b) With the help of relevant diagrams, do the per-phase analysis of three phase connection of transformers for (i) Δ - Δ connection, (ii) Y- Δ connection.	10	L4	4	3,4
6.	a) A transformer bank is made up of three single-phase transformers. The name plate ratings of each single-phase transformer are: 1000 kVA, 13.2/66 kV, $X_l=0.1$ p.u. and $X_m=100$ p.u. Obtain the impedance diagram in per unit values for Y-Y and Δ - Δ interconnections, taking the voltage and power bases for three-phase bank 'induced' by nameplate ratings.	10	L3	4	3

P.T.O.

- b) Define a normal transmission system. Illustrate this with an example. What is the effect that follows when a transmission system is not normal.

10 L4 4 3

Unit – III

7. a) Obtain the equivalent circuit of a three-phase synchronous motor in rotor reference frame.
 b) Derive expressions for mutual inductances between stator and rotor of a three-phase salient-pole synchronous machine considering only one damper winding along the q-axis.
8. a) Obtain the torque equation for a three-phase salient pole synchronous motor in rotor reference frame.
 b) Briefly describe the construction of a permanent magnet synchronous motor. Obtain the model of the above in rotor reference frame.

10 L4 5 3,4,5

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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
